

Talk:

(Arthur Hebecker + Nathaniel Craig)

Outline:

1) SUSY

* Overview of the sol.

* Primer.

* SUSY Breaking (soft)

* MSSM.

2) String Landscape (maybe!).

* Motivation:

1) Only symmetry relating bosons & fermions $\Rightarrow$ logical step towards GUT $\Rightarrow$ provides a path for unification.

2) Only controlled road map from 10d strings to 4D Standard Model involve 4D SUSY.

3) "In principle" resolve hierarchy problem by making Higgs mass technically natural.

• No hierarchy prob if we discard it @ ~ 100 GeV or TeV scale.

• Fine tuning still less severe if we discard it at 10 TeV ($\sim 1\%$).

4) SUSY Theories are much more controllable $\rightarrow$ ADS/CFT

5) Gives possible dark matter candidates.

* Primer (part repeat of Murali).

Poincaré algebra:

$$[P_\mu, P_\nu] = 0, [M_{\mu\nu}, P_\rho], [M_{\mu\nu}, M_{\rho\sigma}] \dots \text{(not imp)}.$$

Any rel QFT must have this $\oplus$ internal symmetries & that's it.
 $d > 1+1$

↑ Coleman-Mandula Thm.

exc: 1) Conformal Symmetry where all particles are massless.

2) Extension to Super Lie Algebra. (unique extension)

Haag-Lopuszanski-Schminers Thm.

→ $\mathbb{Z}_2$ -graded VS $\Rightarrow$ splits into even & odd parts.

New generators are Weyl spinors (instead of bosonic) & the exten looks like

$$\{Q_\alpha, \bar{Q}_{\dot{\alpha}}\} = 2(\sigma^\mu)_{\alpha\dot{\alpha}} P_\mu, \quad \{Q_\alpha, Q_\beta\} = \{\bar{Q}_{\dot{\alpha}}, \bar{Q}_{\dot{\beta}}\} = 0.$$

↑
 $SL(2, \mathbb{C})$ invariant matrices (can use Dirac as well)

$$[P_\mu, Q_\alpha] = 0, \quad [M_{\mu\nu}, Q_\alpha] = i(\sigma_{\mu\nu})_\alpha{}^\beta Q_\beta.$$

$$-\frac{1}{4}(\sigma_\mu \bar{\sigma}_\nu - \sigma_\nu \bar{\sigma}_\mu), (\bar{\sigma}_\mu)_{\dot{\alpha}\dot{\beta}} = \frac{1}{2} \epsilon^{\dot{\alpha}\dot{\beta}} \epsilon^{\mu\nu} (\sigma_\mu)_{\nu\rho} \bar{\sigma}_\nu{}^{\dot{\rho}}{}_{\dot{\beta}}$$

→ full SUSY algebra. $\begin{matrix} \swarrow 4 \text{ bosonic} \\ R^{4|4} \swarrow 4 \text{ real fermionic} \end{matrix}$ 2 $\mathbb{C}$ fermions

* Extension to superspace where in addition to x^μ , there's the fermionic directions θ^α . $\alpha = 1, 2$.

$$\text{anticommuting: } \{\theta^\alpha, \theta^\beta\} = \{\theta^\alpha, \bar{\theta}^{\dot{\alpha}}\} = \{\bar{\theta}^{\dot{\alpha}}, \bar{\theta}^{\dot{\beta}}\} = 0.$$

$$Q_\alpha = \partial_\alpha - i(\sigma^\mu)_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \partial_\mu, \quad \bar{Q}_{\dot{\alpha}} = \bar{\partial}_{\dot{\alpha}} + i\theta^\alpha (\sigma^\mu)_{\alpha\dot{\alpha}} \partial_\mu.$$

Imps called supermultiplets — package together as Superfield
SUSY transf take bosonic dof $\leftrightarrow$ fermionic dof.

Consequences.

(1) # bosonic vs. fermion dof? $\text{Tr}[(t-1)^{N_f}] = 0$ (Witten index).
 $\Rightarrow$ same # fermions & bosons. onshell.

(2) $[P^2, Q_\alpha] = [P^2, \bar{Q}_\alpha] = 0 \Rightarrow$ mass of all particles in a supermultiplet is the same

(3) Except a single extra conserved charge in MSSM $[R, Q_\alpha] = -Q_\alpha$, every other gauge/global generator commutes w/ Q_α $\Rightarrow$ apart from $U(1)_R$ every particle in a supermultiplet has the same Quantum #s.

$\rightarrow$ simply extend a non-SUSY theory by taking any particle we know & put it in a supermultiplet & each supermultiplet has the above 3 properties.

$\rightarrow$ Take any field in SM do the same & a theory w/ unbroken SUSY
Higgs lives in a multiplet w/ a fermion, fermion mass is technically natural b/c of chiral symmetry $\Rightarrow$ Higgs mass is technically natural

$\rightarrow$ So Scalar mass is not technically natural, let's relate it to fermion mass which is technically natural $\Rightarrow$ we've made scalar mass technically natural

* We don't see SM particles & their superpartners w/ the same masses so SUSY needs to be broken — if the symmetry is broken in a controlled way then this is fine, technical naturalness tells us that radiative corrections will be proportional to the breaking scale. So called "soft terms"

Superfields: Build irreps on this enlarged space (super multiplets). $\epsilon^{\alpha\beta} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

$$F(x, \theta, \bar{\theta}) = f(x) + \theta^\alpha \varphi_\alpha(x) + \bar{\theta}^{\dot{\alpha}} \bar{\chi}_{\dot{\alpha}}(x) + \theta^2 m(x) + \bar{\theta}^2 n(x) + \theta^\alpha \sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} v_\mu(x) + \theta^2 \bar{\theta}^2 d(x) + \bar{\theta}^2 \theta^\alpha \psi_\alpha(x) + \theta^2 \bar{\theta}^2 \lambda(x)$$

maybe say this & don't write.

$$\delta_\xi F = (\xi Q + \bar{\xi} \bar{Q}) F$$

Too many dof, simplify (define sub-reps). (offshell v. onshell). Auxiliary field

1) Chiral Superfield $\rightarrow$ No antichiral dep. $\Phi = \varphi(y) + \sqrt{2} \theta^\alpha \psi_\alpha(y) + \theta^2 F(y)$

$$\bar{D}_{\dot{\alpha}} \Phi = 0 \quad w/ \quad \bar{D}_{\dot{\alpha}} = -\bar{\partial}_{\dot{\alpha}} - i \theta^\alpha (\sigma^\mu)_{\alpha\dot{\alpha}} \partial_\mu$$

$y^\mu = x^\mu + i \theta \sigma^\mu \bar{\theta}$
shifted field

$\rightarrow$ Can also define an antichiral one.

2) Real Superfield $\rightarrow$ equal to its complex conjugate.

$$V = \bar{V}$$

$$V = -\theta \sigma^\mu \bar{\theta} A_\mu + i \theta^2 \bar{\theta} \bar{\lambda} - i \bar{\theta}^2 \theta \lambda + \frac{1}{2} \theta^2 \bar{\theta}^2 D$$

SUSY invariant Lagrangian

1) Chiral Superfields Lagrangian

$$\int d\theta d\bar{\theta} = 1 \rightarrow \text{zero dim}$$

$$\mathcal{L} = \int d^4\theta d^2\bar{\theta} K(\Phi^i, \bar{\Phi}^{\dot{i}}) + \int d^2\theta W(\Phi^i) + \text{h.c.}$$

$\uparrow$
called Kahler potential

$\uparrow$
superpotential
chiral superfield

* We need to pick out the top component of a representation of the superfield st the SUSY transf of the field is a total derivative & the action is SUSY invariant.

eg. Wess-Zumino Model — simplest possible non-trivial SUSY model.

$$K = \bar{\Phi}\Phi, \quad W = \frac{m\Phi^2}{2} + \frac{\lambda}{3}\Phi^3.$$

$$\mathcal{L} = -|\partial\varphi|^2 - i\bar{\Psi}\bar{\sigma}^\mu\partial_\mu\Psi - \left(\frac{m}{2}\psi^2 + \lambda\psi^3\right) + h.c. - \underbrace{|m\varphi + \lambda\varphi^2|^2}_{|F|^2} \quad \left. \vphantom{\mathcal{L}} \right\} \text{No need to write this}$$

2) Real Superfields Lagrangian.

$$\mathcal{W}_\alpha = -\frac{1}{4}\bar{D}^2 D_\alpha V.$$

$$\mathcal{L} = \frac{1}{2g^2} \text{tr} \left(\omega^2|_{\theta^2} + h.c. \right) + \bar{\Phi}^+ e^{2V} \Phi|_{\theta^2\bar{\theta}^2} + W(\Phi)|_{\theta^2} + h.c.$$

under super gauge transf: $e^{2V} \rightarrow e^{\Lambda^+} e^{2V} e^\Lambda, \quad \Phi \rightarrow e^{-\Lambda} \Phi.$

So Off-shell Lagrangian:

$$\mathcal{L} = \frac{1}{g^2} \text{tr} \left\{ -\frac{1}{2} F_{\mu\nu} F^{\mu\nu} - 2i\bar{\lambda}\bar{\sigma}^\mu D_\mu \lambda + D^2 \right\} - |\partial_\mu\varphi|^2 - i\bar{\Psi}\bar{\sigma}^\mu D_\mu\Psi + |F|^2 + i\sqrt{2}(\varphi^\dagger\lambda\Psi - \bar{\Psi}\bar{\lambda}\varphi) + \varphi^\dagger D\varphi.$$

* SUSY Breaking:

Q mixes fermions & bosons & $[Q_\alpha, P^\mu p_\mu] = 0 \Rightarrow$ fermions & bosons have the same mass in a superfield / supermultiplet.

→ SUSY needs to be broken in the real world.

→ vacuum should not be SUSY invariant.

2 simplest ways, give either F or D a non-zero vev.

1) F term breaking $\mathcal{L} \supset \bar{F}F + cF + h.c$

$$\Rightarrow \bar{F} = -c \neq 0 \text{ in vacuum}$$

Wess-Zumino does not have SUSY breaking since the linear term can be absorbed into a shift of Φ . We actually need a model made out of atleast 3 chiral superfields.

"O'Raifeartaigh Model".

2) D term breaking $\mathcal{L} = \frac{1}{2g^2} W^2|_{\theta^2} + 2KV|_{\bar{\theta}^2\theta^2}.$

$$\supset \frac{1}{2g^2} D^2 + 2D \Rightarrow D = -Kg^2 \neq 0$$

in vacuum.

"Fayet - Iliopoulos" term

→ Can then easily expand around the vev to find spontaneous SUSY breaking.

* Minimal SUSY SM — MSSM

* SM fermions + scalars → Chiral Superfields.

* " vectors → real superfields.

* Higgs goes to 2 chiral superfields, so the full set is

$$\Phi^a = \{Q, U, D, L, E, H_u, H_d\}.$$

real superpotential

$$\mathcal{L}_{\text{gauge}} = \sum_i^3 \text{tr} \left(\frac{1}{2g_i^2} (w_i)^\alpha (w_i)_\alpha \right) |_{\theta^2} + h.c.$$

Kahler

$$\mathcal{L}_K = \sum_{a=1}^7 \Phi_a^\dagger e^{2V} \Phi_a |_{\theta^2 \bar{\theta}^2} \quad \text{w/} \quad V = V_1 + V_2 + V_3 \quad - 3 \text{ real superfields}$$

chiral
superpotential

$$\mathcal{L}_W = (W_\mu + W_\gamma + W_e)|_{\theta^2} + \text{h.c.}$$

Higgs $\hookrightarrow W_\mu = \mu H_u H_d \leftarrow \text{problematic (will discuss later)} \rightarrow \mu \text{ problem}$

Yukawa $\hookrightarrow W_\gamma = \lambda_u Q H_u U + \lambda_d Q H_d D + \lambda_e L H_d E$

Extra $\hookrightarrow W_e = a L H_u + b Q L D + c U D + d L L E$

- 1) We see why we needed $H_u, H_d \rightarrow$ otherwise we won't have mass for all fermions through Yukawa.
- 2) Also need to cancel the $U(1)$ gauge anomaly introduced by the Higgsino in H_d .

imp

We introduce proton decay & lepton # violation $\rightarrow$ forbid these.
We have also truncated to cubic order in W st our theory is renormalizable $\rightarrow$ mass dim 3 in W corresponds to mass dim 4 in $\mathcal{L}$ since prop onto θ^2 raises mass dim by 1.

To forbid W_e we can introduce a global $U(1)_R$ symmetry that would forbid these but we lose the W_μ term as well so we restrict to just the $Z_2 = \{1, -1\} \subset U(1)_R$, transformation rules under Z_2 R-parity are:
Even: Higgs scalar, fermions, gauge bosons.
Odd: Higgsino, sfermion, gauginos.

EWSB w/ W_μ becomes awkward.

$\Rightarrow$ Also implies that the odd superpartners cannot decay into even standard model particles $\Rightarrow$ stable $\Rightarrow$ DM candidates.

* Let's arrive at MSSM using spontaneous SUSY breaking

Toy Model. $\mathcal{L} = (\bar{S}S - c(\bar{S}S)^2)|_{\theta^2\bar{\theta}^2} + c_2 S|_{\theta^2} + \text{h.c.} + \bar{\Phi}\Phi|_{\theta^2\bar{\theta}^2} + \frac{1}{M^2} \bar{\Phi}\Phi \bar{S}S|_{\theta^2\bar{\theta}^2}$

S, Φ CSF but $S_F \neq 0 \Rightarrow S$ has spon SUSY breaking

Φ free massless chiral superfield (corresponds to SM)

- Last term higher dim op suppressed by large M coupling the two sectors.

$$\mathcal{L} \supset \Phi \Phi |_{\theta^2 \bar{\theta}^2} + \frac{1}{M^2} \bar{\Psi} \Psi \bar{F}_S F_S \rightarrow m_\Psi = \frac{\bar{F}_S F_S}{M^2}$$

- * Can also introduce a SUSY invariant mass term in superpotential since Non-renormalization theorem tells us that this would not be subject to renormalization.

Let's take the easy way out

Soft SUSY Breaking occurs b/c of Spontaneous Superfield (hidden sector) S into SUSY SM. 4 types of higher dim terms that generate all SUSY breaking effects.

$$\mathcal{L}_1 = \frac{1}{M^2} \bar{\Phi} \Phi \bar{S} S |_{\theta^2 \bar{\theta}^2} \rightarrow \text{sparticle soft mass } |M_0|^2 \phi^2$$

$$\mathcal{L}_2 = \frac{1}{M^2} \Phi^2 \bar{S} S |_{\theta^2 \bar{\theta}^2} \rightarrow \text{holomorphic soft mass } B_\mu$$

$$\mathcal{L}_3 = \frac{1}{M} \Phi^3 S |_{\theta^2} + \text{h.c.} \rightarrow \text{higgsino trilinear interaction}$$

$$\mathcal{L}_4 = \frac{1}{M} W^\alpha W_\alpha S |_{\theta^2} + \text{h.c.} \quad \text{eg: } \Phi^3 \sim Q H_u U$$

↓
gaugino mass, $M_{1/2}$

$$\text{If } M \sim M_P \sim 10^{18} \text{ GeV} \rightarrow M_0 \sim A \sim M_{1/2} \sim 1 \text{ TeV.}$$

↑
SUSY Breaking
Scale
needed.

SUGRA
Can also have a $\frac{1}{M} \Phi^2 S |_{\theta^2}$ term which maybe a problem but there is an elegant sol to that Giudice-Masiero mechanism.

μ=0 prob is the before EWSB, Higgsinos are massless & after it kicks in they are still very light so we should've seen it.

+ EWSB condition becomes less flexible.

SUGRA

μ=0 otherwise the natural scale is 10^{10} GeV . Impose global $U(1)_R$ to forbid it.

$$K_{\text{term}} \sim \frac{S H_u H_d}{M}$$

$$M_{\text{eff}} \sim \sqrt{B_\mu} \sim \frac{F_S}{M} \sim m_{\text{eff}}$$

Pure SUSY sol involve introducing
1) a singlet field
2) another superpotential spurion.

NMSSM

~ Higgs mass
at 10^6 GeV

which forms it h.c. of subsequent EWSB.

$$\rightarrow M_H^2 \propto \underbrace{\mu^2}_{\text{SUSY}} + \underbrace{M_{H_u}^2 + m_{H_d}^2}_{\text{SUSY}} + \underbrace{\text{soft bilinear terms}}_{\text{tree}} + \text{SUSY breaking (after the mass of stop + gaugino + A term)} + \frac{6}{16\pi^2} y_t^2 (m_{t_1}^2 + m_{t_2}^2) - \frac{6y_t^2}{16\pi^2} \Lambda^2 + \frac{6y_t^2}{16\pi^2} \Lambda^2$$

$m_{\text{soft}}^2 \log\left(\frac{\Lambda}{m_{\text{soft}}}\right)$

↑ UV sensitivity goes away.

- All parameters (a lot more than ordinary SM) are now protected from power law divergences.
- EWSB can proceed as regime
- SUSY Breaking & weak scales are naturally small
- Model is renormalizable can be valid upto Planck scale
- Predict $M_{\text{GUT}} \sim 10^{16}$ GeV

* Probs:

- 1) Higgs mass is bounded by Z-boson mass at tree level
→ not true. Correction needed to bring it upto 125 GeV at loop level requires a large stop mass, drives the Higgs vev to higher values, partially spoils hierarchy prob resolution
- 2) Would've been the best sol 20 years ago until LHC
No SUSY and 10 TeV so MCSM may not be a weak scale model but a model at sig higher scale $m_{\text{soft}} \gg m_{\text{ew}}$

Reqs Higgs mass matrix lower eigenvalue (res for EWSB) is smaller than the typical entries so fine tuning of order $\frac{m_{\text{ew}}^2}{m_{\text{soft}}^2} \sim 1\%$ level.
 $m_{\text{soft}} \sim \text{at least } 10 \text{ TeV or above} \rightarrow \text{little hierarchy prob.}$

String Landscape:

- Very diff attitude now that emerges from String Theory called the landscape.
Hienly problem is not removed by realized as a special choice among a large # of possible vacua.

- * One UV theory

- * Many 4D vacua

- * Parameters like Λ & m_H^2 scan over this landscape.

1) Fundamental theory.

- In 10d, 5 string theories which are low energy limits of 11D M-theory

- To get 4d, you compactify 6 extra dims on some curved space
(called a Calabi-Yau Manifold).

- Choices.

- * 5 string theory.

- * Type of 6D geometry (which Calabi-Yau Manifold).

- * Branes, orientifolds planes & fluxes.

- gives diff sets of low energy EFT $\sim 10^{272000}$ distinct flux vacua
(rough modern estimate)

- Possible that many have a MSSM-like gauge group & matter content
+ positive & small cosmological constant.

- So 1 UV theory → large menu of 4d low-energy worlds many of which may look like ours.

- No saturation and $m_H^2 \sim 0$. Natural scale is $m_H^2 \sim m_p^2$

Higgsing $\sim 10^{-32}$, $\Lambda \sim 10^{-120}$. Even w/ only 10^{600} possible vacua, a tiny by

$10^{-32} \times 10^{-120}$ can be realized by picking a rare enough vacuum.

- So in the landscape this thing is not forbidden but for reasons unknown we live in a very small corner of a high discrete set of vaks.
- Anthropic argument: among all vacua, possibly only a small spec^l # permit long-lived structures, chemistry & observers.
- What fraction of vaks live in low-scale vs. high-scale SUSY. Eternal inflation can populate all these so may be able to give us an answer but we don't have one rn. (kind of measurement problem).